

Bachelor in Computer Applications

Course Title: **Information Security**

Code No: Semester: **5**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. Define Euler totient function with an example. Find the GCD of 12 and 32 using Extended Euclidean algorithm.
2. Describe the role of hash functions in authenticating message? How SHA - 1 algorithm is used to produce hash value of a message? Explain.
3. Distinguish between threats and attack. Discuss some computer security strategy. Assume a prime number 23 and 9 as its primitive root. Alice select a private key 5 and Bob select the private key 6. Now find the secret key value that Alice and Bob shared using Diffie - Hellman protocol.
4. Discuss about two factor authentication with an example.
5. Explain the different types of access control principles.
6. Describe any two types of malicious software.
7. What is risk assessment? Describe the security auditing architecture.
8. Do cybercrime and computer crimes refers to same? Justify with relevant scenarios.
9. What do you mean by countermeasures for malwares? Discuss about audit trail analysis.
10. List some issues for user authentication. What is trust framework?
11. What is the role of digital signature in message authentication? List any two natures of zombies.
12. Write the algorithm for RSA key generation with encryption and decryption.